

Student & Hardware Metadata Field	Laboratory Session Specifications & Target Machine Declaration
Student Full Name	Seidaly Burkit
Student Identification Number	230103108
Date & Laboratory Session	18.09.2026 10:20
Processor SKU / SoC Model	i5-12450H
Core Topology (Physical P-cores / E-cores)	12
Cache Hierarchy Capacity	704 Kb 7Mb 12Mb

Task 1

The results :

```
Run task1_amdahl x
D:\PYY\python.exe D:\pm_labs\task1_amdahl.py
OS Reported Logical Cores: 12
Single Process Baseline Time T(1): 7.3979s

Cores | Time(s) | Observed Speedup | Theoretical Linear
-----|-----|-----|-----
1 | 3.3488 | 2.21x | 1.00x
2 | 2.1154 | 3.50x | 2.00x
4 | 1.3814 | 5.36x | 4.00x
8 | 1.1636 | 6.36x | 8.00x
12 | 1.1599 | 6.38x | 12.00x
16 | 1.4149 | 5.23x | 16.00x

Process finished with exit code 0
```

The result table :

Worker Cores (p)	Measured Time T(p)	Observed Speedup S(p)	Ideal Linear	Parallel Efficiency
1	7.3979 s	1.00x	1.00x	100.0%
2	2.1154 s	3.50x	2.00x	175.0%
4	1.3814 s	5.36x	4.00x	134.0%
8	1.1636 s	6.36x	8.00x	79.5%
12	1.1599 s	6.38x	12.00x	53.2%

16

1.4149 s

5.23x

16.00x

32.7%

Q1.1

The inflection point is $p = 12 \text{ cores}^*$. Speedup increases from 1 to 12 cores (6.38 $\times$), but degrades at 16 cores (5.23 $\times$). My system reports 12 logical cores, so 16 workers exceed the available logical CPU capacity, causing resource contention and multiprocessing overhead.

Q1.2

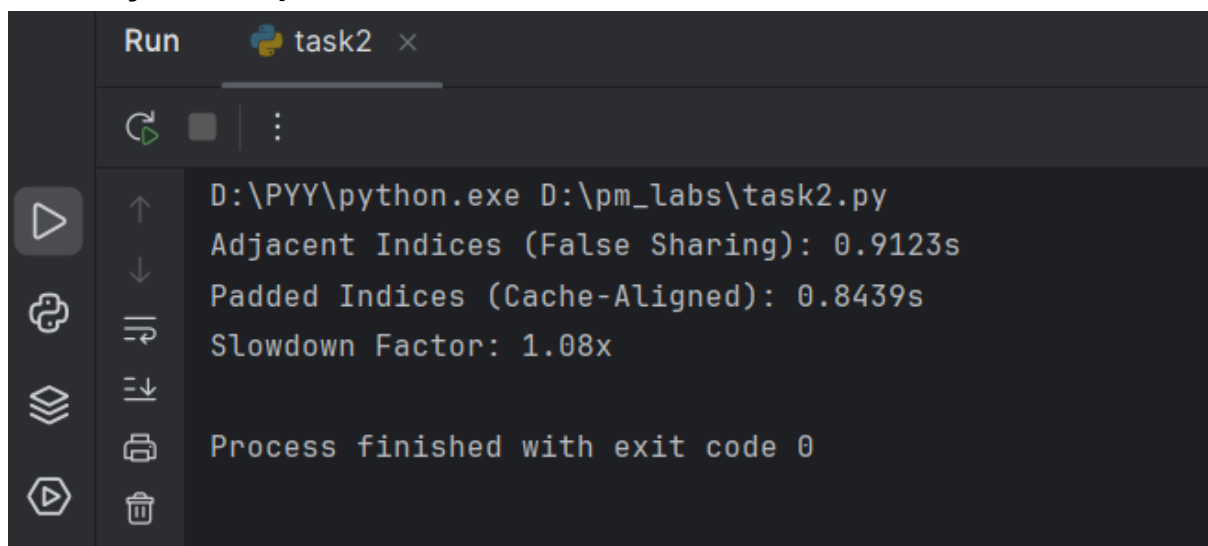
$P \approx 108.4\%$. This is physically impossible, indicating measurement noise/benchmark overhead because the single-process baseline was much slower than the later $p=1$ run.

Q1.3

This negative result is not physically meaningful. It confirms that the measured baseline is inconsistent and cannot be used to derive a valid Amdahl limit. A 128-core server would therefore not be expected to achieve a meaningful 128 $\times$ speedup from these measurements alone.

Task 2

In totally 3 attempts:



```
Run task2 x
D:\PYY\python.exe D:\pm_labs\task2.py
Adjacent Indices (False Sharing): 0.9123s
Padded Indices (Cache-Aligned): 0.8439s
Slowdown Factor: 1.08x
Process finished with exit code 0
```

```
Run task2 x
D:\PYY\python.exe D:\pm_labs\task2.py
Adjacent Indices (False Sharing): 1.0700s
Padded Indices (Cache-Aligned): 0.9748s
Slowdown Factor: 1.10x
Process finished with exit code 0
```

```
Run task2 x
D:\PYY\python.exe D:\pm_labs\task2.py
Adjacent Indices (False Sharing): 0.8860s
Padded Indices (Cache-Aligned): 0.7859s
Slowdown Factor: 1.13x
Process finished with exit code 0
```

The result table :

Memory Layout Configuration	Trial 1 [s]	Trial 2 [s]	Trial 3 [s]	Median Time [s]
Adjacent Indices (False Sharing Active)	0.9123	1.0700	0.8860	0.9561
Padded Indices (Cache-Line Aligned)	0.8439	0.9748	0.7859	0.8682